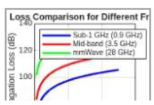
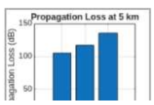
10:42

History

```
clc;
clear;
close all;
% Frequency Bands (GHz)
f_sub = 0.9;        % Sub-1 GHz
f_mid = 3.5;        % Mid-band
f_mm = 28;          % mmWave
% Communication Distance (km)
d = 0.1:0.1:5;
% Free Space Path Loss (FSPL)
PL_sub = 20*log10(d) + 20*log10(f_sub*1000) +
32.44;
PL_mid = 20*log10(d) + 20*log10(f_mid*1000) +
32.44;
PL_mm = 20*log10(d) + 20*log10(f_mm*1000) +
32.44;
% Display Sample Results
disp('Propagation Loss Analysis');
disp('----------------------------------------
------------------');
fprintf('Distance(km)\tSub-1GHz(dB)
\tMid-band(dB)\tmmWave(dB)\n');
for i = 1:length(d)
    fprintf('%6.1f\t\t%8.2f\t\t%8.2f\t\t%8.2f\
n', ...
        d(i), PL_sub(i), PL_mid(i), PL_mm(i));
end
% Plot Propagation Loss
figure;
plot(d, PL_sub,'b','LineWidth',2);
hold on;
plot(d, PL_mid,'r','LineWidth',2);
plot(d, PL_mm,'g','LineWidth',2);
grid on;
xlabel('Communication Distance (km)');
ylabel('Propagation Loss (dB)');
title('Propagation Loss Comparison for
Different Frequency Bands');
legend('Sub-1 GHz (0.9 GHz)','Mid-band
(3.5 GHz)','mmWave (28
GHz)','Location','northwest');
% Bar Graph at 5 km
figure;
Loss = [PL_sub(end) PL_mid(end) PL_mm(end)];
bar(Loss);
set(gca,'XTickLabel',{'Sub-1
GHz','Mid-band','mmWave'});
ylabel('Propagation Loss (dB)');
title('Propagation Loss at 5 km');
grid on;
```





10:42

Propagation Loss Analysis

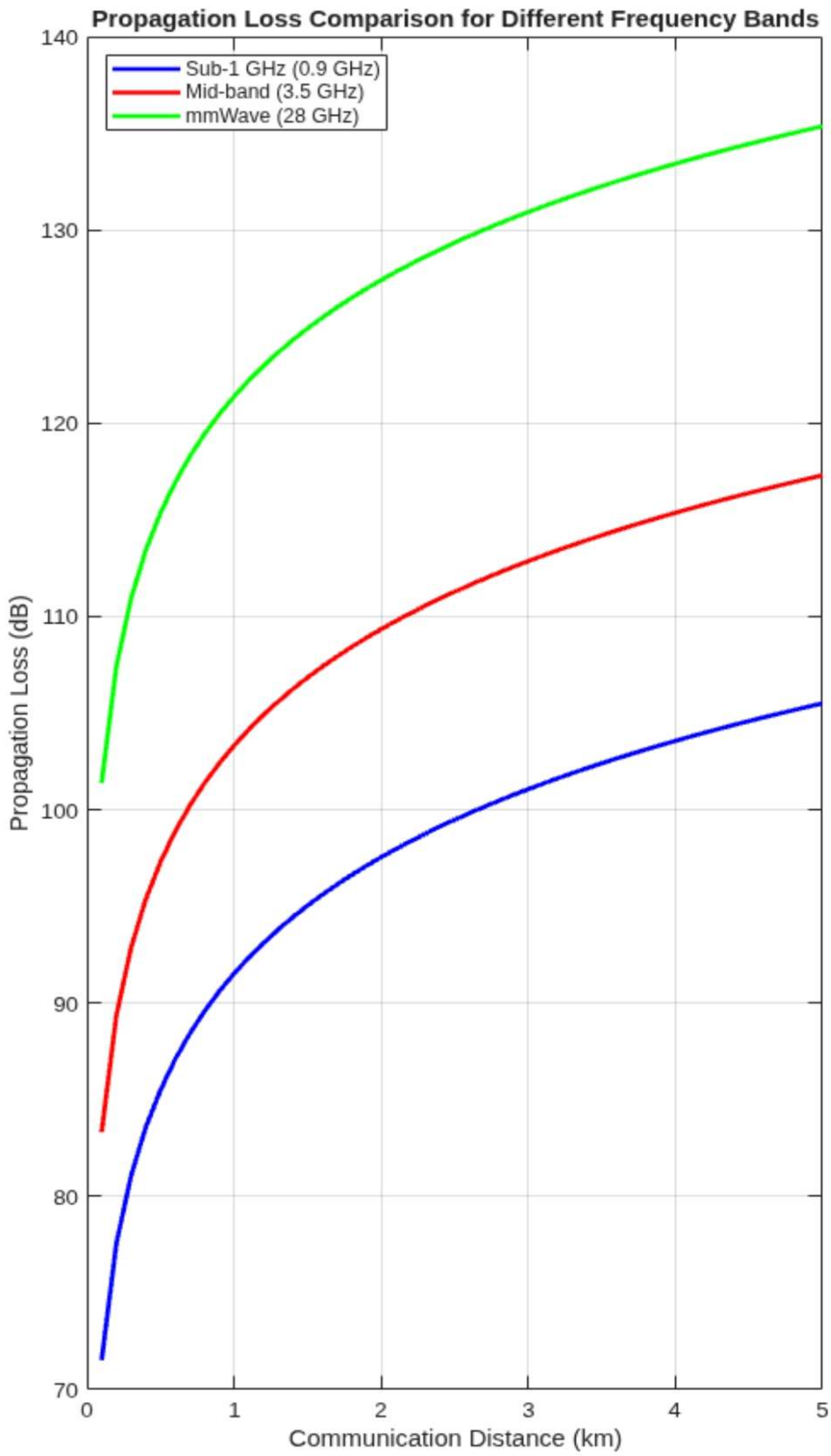
Distance(km)	Sub-1GHz(dB)	Mid-band(dB)	mmWave(dB)
0.1	71.52	83.32	101.38
0.2	77.55	89.34	107.40
0.3	81.07	92.86	110.93
0.4	83.57	95.36	113.42
0.5	85.50	97.30	115.36
0.6	87.09	98.88	116.95
0.7	88.43	100.22	118.29
0.8	89.59	101.38	119.44
0.9	90.61	102.41	120.47
1.0	91.52	103.32	121.38
1.1	92.35	104.15	122.21
1.2	93.11	104.90	122.97
1.3	93.80	105.60	123.66
1.4	94.45	106.24	124.31
1.5	95.05	106.84	124.90
1.6	95.61	107.40	125.47
1.7	96.13	107.93	125.99
1.8	96.63	108.43	126.49
1.9	97.10	108.90	126.96
2.0	97.55	109.34	127.40
2.1	97.97	109.77	127.83
2.2	98.37	110.17	128.23
2.3	98.76	110.56	128.62
2.4	99.13	110.93	128.99
2.5	99.48	111.28	129.34
2.6	99.82	111.62	129.68
2.7	100.15	111.95	130.01
2.8	100.47	112.26	130.33
2.9	100.77	112.57	130.63
3.0	101.07	112.86	130.93
3.1	101.35	113.15	131.21
3.2	101.63	113.42	131.49
3.3	101.90	113.69	131.75
3.4	102.15	113.95	132.01
3.5	102.41	114.20	132.26
3.6	102.65	114.45	132.51
3.7	102.89	114.69	132.75
3.8	103.12	114.92	132.98
3.9	103.35	115.14	133.20
4.0	103.57	115.36	133.42
4.1	103.78	115.58	133.64
4.2	103.99	115.79	133.85
4.3	104.19	115.99	134.05
4.4	104.39	116.19	134.25
4.5	104.59	116.39	134.45
4.6	104.78	116.58	134.64
4.7	104.97	116.76	134.83
4.8	105.15	116.95	135.01

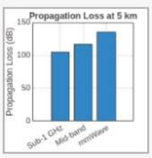
>> Enter command

10:42

Figures SHOW DATA CURSOR

Figure 1: clc; clear; close all; % Frequency Bands (GHz) f_sub = 0.⋯





10:42

Figures

SHOW DATA CURSOR

Figure 2: clc; clear; close all; % Frequency Bands (GHz) f_sub = 0⋯